Specific Aims

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More than 90% of cancer-related deaths are caused by drug resistance in patients receiving targeted therapy[1], but we still do not have a framework that predicts resistance by reasoning over the causal hierarchy that shows how conflicting multi-omics signals resolve in the living cell. GDSC2 and CCLE/DepMap [2] have mapped multiomics profiles of cancer cell line-drug pairs that are publicly available. Deep learning baselines (DRN-CDR [3], MSDRP[4]) resolve conflicting data layers using learned weights that are opaque and uninterpretable by clinicians. Natural-language reasoning is possible today using LLMs but they suffer from context overwhelm and sycophantic convergence without explicit biological rules [5]. The critical gap is the absence of a framework that uses the causal ordering of evidence layers (DNA structural state, transcriptional activity, pathway topology and pharmacological history) as an executable logic for resolving agentic conflicts. Without this, oncology AI cannot have robust predictions that a clinician can inspect and contest, eliminating accuracy-optimized black boxes and relying on accountable clinical decision support.

The **long term goal** is to design a multi-modal augmentation as the new way for oncology AI that is biologically faithful. The **objective** is to build, evaluate and ablate a decentralized multi-agent consensus framework for predicting cancer drug resistance which is governed by a formal biological Hierarchy of Truth with a full debate trace. *We hypothesize that a 4-agent system (Genomics, Transcriptomics, Pathways and Pharmacology) resolving conflicts using the 5-tier axiom hierarchy will predict drug resistance more accurately with faithful reasoning than monolithic LLMs and naive majority-voting multi-agent systems.* This is grounded in: (1) published tumor board practices and (2) technical feasibility confirmed by implementing a smaller prototype with the conflict resolution protocol and validated with a few real examples. Completion of this study will shift the evaluation of oncology AI to reasoning quality (hallucination rate and faithfulness scores) which are absent from existing work. My guide and I are well-prepared for this study owing to our prior work with MCPs, agentic AI and multiomics data analysis on cancer datasets.

Specific Aim 1: Determine whether axiom-governed multi-agent debates predict cancer drug resistance more accurately than monolithic LLMs and classical ML baselines.
*Working hypothesis:*Causal tier resolution prevents conflicts from collapsing into statistical averaging, thus improving AUROC, AUPRC from a single LLM and RandomForest baselines trained on the same data.
*Approach:*Run the full system and the baselines on 40 GDSC2 development cases across different LLMs and validate on held-out CTRPv2 cases.
*Milestones/metrics:*AUROC ≥ 0.75 and AUPRC ≥ 0.70 for the system, aim for the framework to exceed both baselines on the metrics.

Specific Aim 2: Determine which architectural components independently contribute to predictive accuracy and reasoning quality through ablation studies.
*Working hypothesis:*Excluding axiom hierarchy, using single agent or replacing MCPs returns free-text summaries which will independently lower performance confirming that these are nonredundant components.
*Approach:*Execute six ablations (no axioms, no debate, single agent, free-text tools, randomized tier order, majority-vote) on the 40 test cases.
*Milestones/metrics:*Each ablation shows significant degradation on ≥ 1 of: AUROC, hallucination rates, or the faithfulness score.

The proposed study is innovative because it sets the foundation to formalize inter-modality conflict resolution in oncology AI as executable axioms [5], uses MCPs as an anti-hallucination layer [7, 8] and uses reasoning quality metrics as the primary evaluation metrics with predictive accuracy. We expect the framework to perform better than all the baselines on AUROC and AUPRC and achieve a hallucination rate below 5%, and for each ablation to independently lower performance. These outcomes will establish an open-source, model-agnostic framework that can be deployed for clinicians and researchers to generate auditable drug sensitivity predictions. This also advances the mission of NCI to reduce cancer mortality though accountable AI-assisted precision oncology.

Reflection, Feedback & References

Reflection

The discipline of compressing the specific aims to one page made us shift the way we think of framing the objective and the multiple hypotheses statements that were present in the short proposal document. The hypothesis had to be simple enough for anyone to understand it but also should contain the core idea of the study. Drafting the document to be more conceptual rather than procedural was the most productive approach.

Feedback

Feedback was collected from the first progress meeting from Dr. Lessick about the inclusion of PPI networks as one of the agents instead of the KEGG pathways agent. This is a very valuable advice received and we did anticipate such feedback given that there are multiple databases available that could be better suited to the use case being addressed. Keeping this in mind, we are designing the framework in such a way that models can be desiged and plugged into the framework without having to disrupt the entire structure. This way with the evolution of better datasets, we can always replace or add agents as we need and the predictions will just get more fine-tuned.

References

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